2.5 Krull Dimension

The purpose of this section is to abstract the notions of Krull Dimension in commutative ring theory and topology. A more standard approach (eg EGA IV) would be to develop the topological notion first, and then link to commutative ring case using the prime spectrum. Generally the concept is not well-behaved, so stronger conditions are defined which generally hold in geometric cases. Principle references are (EGA0 IV 14.3, Heinrich).

Definition 2.5.1 (Finite-Dimensional Poset)
*Let $(\mathcal{G}, \leq)$ be a poset, we say that it is **finite-dimensional** if*

$$\dim(\mathcal{G}) := \sup\{\ell(C) \mid C \subseteq \mathcal{G} \text{ a chain}\} < \infty$$

In this case we define

$$\begin{aligned}\dim(x) &:= \dim\big(\{x\}^{\downarrow}\big)\\ \operatorname{codim}(y) &:= \dim\big(\{y\}^{\uparrow}\big)\\ \operatorname{codim}(y, x) &:= \dim\big(\{x\}^{\downarrow} \cap \{y\}^{\uparrow}\big)\end{aligned}$$

Note $\mathcal{G}$ is both Noetherian and Artinian, but finite-dimensionality is a stronger condition. Note also that $\{x\}^{\downarrow}, \{y\}^{\uparrow}$ and $\{x\}^{\downarrow} \cap \{y\}^{\uparrow}$ are finite-dimensional posets

Definition 2.5.2 (Krull Lattice)
*Let $(\mathcal{F}, \leq)$ be an **Artinian** distributive lattice. We say it is a **Krull Lattice** if the poset of join-prime elements $\mathcal{J}(\mathcal{F})$ is finite-dimensional and define*

$$\dim(\mathcal{F}) := \dim(\mathcal{J}(\mathcal{F}))$$

By (2.4.3) we have $\mathcal{H} = \{x\}^{\downarrow}, \{y\}^{\uparrow}, \{x\}^{\downarrow} \cap \{y\}^{\uparrow}$ are Krull Lattices such that

$$\mathcal{J}(\mathcal{H}) = \mathcal{J}(\mathcal{F}) \cap \mathcal{H}$$

For $x, y \in \mathcal{F}$ we have a unique decomposition into maximal join-prime elements $x_i, y_j \in \mathcal{J}(\mathcal{F})$ (2.4.7)

$$x = x_1 \vee \ldots \vee x_n$$

$$y = y_1 \vee \ldots \vee y_m$$

Note that $y \leq x \iff$ for all j we have $y_j \leq x_i$ for some i and we may define

$$\begin{aligned}\dim(x) &:= \max_i \dim(x_i) = \dim\big(\{x\}^{\downarrow}\big)\\ \operatorname{codim}(y) &:= \min_j \operatorname{codim}\big(y_j\big)\\ \operatorname{codim}(y, x) &:= \min_j \max_i \big\{\operatorname{codim}\big(y_j, x_i\big) \mid y_j \leq x_i\big\}\\ &= \min_j \operatorname{codim}\big(y_j, x\big)\end{aligned}$$

note it's required to be careful in definition of co-dimension in order to have a sensible co-dimension formula. Note also that

$$\dim\big(y; \{x\}^{\downarrow}\big) = \dim(y)$$

Remark 2.5.3
For the topological case, we would define $\mathcal{F}$ to be the closed subsets of X and $\mathcal{J}(\mathcal{F})$ would be the collection of irreducible closed subsets.

Proposition 2.5.4 (Extending chains)
Let $(\mathcal{G}, \leq)$ be a finite-dimensional poset

- *Every chain is contained in a saturated chain*
- *Every chain is contained in a maximal chain*
- *Every maximal chain is of the form*

  $$x_0 \lneq x_1 \ldots \lneq x_n$$

  for x_0 minimal and x_n maximal in $\mathcal{G}$.

Definition 2.5.5 (Properties)
Let $\mathcal{G}$ be a finite-dimensional poset. Then we say it is

- ***Irreducible*** *if it has a top element*
- ***Equidimensional*** *if every maximal element has the same dimension*
- ***Equicodimensional*** *if every minimal element has the same codimension*
- ***Biequidimensional*** *if every maximal chain has the same length*
- ***Quasi-Biequidimensional*** *if $\{x\}^{\downarrow}$ is biequidimensional for all x (equivalently all maximal x)*
- ***Catenary*** *if for every pair $y \leq x$, every saturated chain in $[y,x] := \{y\}^{\uparrow} \cap \{x\}^{\downarrow}$ has the same length, namely* $\operatorname{codim}(y,x)$.

If $\mathcal{F}$ is a Krull Lattice then we say it inherits these properties from $\mathcal{J}(\mathcal{F})$. Note if $\mathcal{F}$ is irreducible then it also has (the same) top element.

Trivially irreducible implies equidimensional. Similarly biequidimensional implies both equidimensional and equicodimensional, but not conversely.

Finally ***quasi-biequidimensional + equidimensional $\iff$ biequidimensional.***

Proposition 2.5.6 (Simple Properties)
Let $\mathcal{G}$ be a finite-dimensional poset then

a) If x is maximal then $\operatorname{codim}(x) = 0$
b) If x is minimal then $\dim(x) = 0$
c) $\dim(\mathcal{G}) = \sup\{\dim(x) \mid x \text{ maximal}\} = \sup\{\operatorname{codim}(x) \mid x \text{ minimal}\}$
d) For all $z \leq y \leq x$ we have $\operatorname{codim}(z,y) + \operatorname{codim}(y,x) \leq \operatorname{codim}(z,x)$

If $\mathcal{F}$ is a Krull Lattice then

e) For all $y \leq x$ we have $\dim(y) + \operatorname{codim}(y,x) \leq \dim(x)$
f) For all $y \leq x$ we have $\operatorname{codim}(y,x) = 0 \iff y_j = x_i$ *some i,j*

Alternatively $\operatorname{codim}(y,x) > 0$ *if and only if* $\big(y_j \leq x_i \implies y_j \neq x_i\big)$.

Proof. e) The case of a finite-dimensional poset is (relatively) clear. In the general case then we have

$$\dim\big(y_j\big) + \operatorname{codim}(y,x) \leq \dim\big(y_j\big) + \operatorname{codim}\big(y_j,x\big) = \max_i\big(\dim\big(y_j\big) + \operatorname{codim}\big(y_j,x_i\big)\big) \leq \max_i \dim(x_i) = \dim(x)$$

and taking max over j yields the result.

f) The case $x,y \in \mathcal{J}(\mathcal{F})$ is clear by (2.5.4). For the general case $\operatorname{codim}(y,x) = 0 \iff \operatorname{codim}\big(y_j,x\big) = 0$ for some $j \iff \big(y_j \leq x_i \implies \operatorname{codim}\big(y_j,x_i\big) = 0\big) \iff y_j = x_i$ for some i,j. ∎

Corollary 2.5.7 (Codimension 1 formula)
Let $\mathcal{F}$ be a Krull Lattice with $y \in \mathcal{F}$, $x \in \mathcal{J}(\mathcal{F})$ and $y \leq x$. Then

$$\dim(y) = \dim(x) - 1 \implies \operatorname{codim}(y,x) = 1$$

Suppose further that $\mathcal{F}$ is irreducible then

$$\dim(y) = \dim(\mathcal{F}) - 1 \implies \operatorname{codim}(y) = 1$$

Proof. By (2.5.6) e) $\operatorname{codim}(y,x) \leq 1$. If $\operatorname{codim}(y,x) = 0$ then by f) we see that $y_j = x$ for some j, whence $y = x$ which contradicts $\dim(y) = \dim(x) - 1$. Therefore $\operatorname{codim}(y,x) = 1$ as required.

If $\mathcal{F}$ is irreducible the result follows with $x = \top$. ∎

Remark 2.5.8 (Duality)
We note that the concepts of ***dimension*** *(of a poset).* ***biequidimensional*** *and* ***catenary*** *are* self-dual, *in the sense that they are preserved when considering the dual poset* $\left(\mathcal{G}, \overset{d}{\leq}\right)$.

Similarly the concepts of ***equidimensional*** *and* ***equicodimensional*** *are dual to each other.*

Proposition 2.5.9
Let $\mathcal{G}$ be a finite-dimensional poset. Then the following are equivalent

- *$\mathcal{G}$ is catenary*

- *For every triplet $z < y < x$ in $\mathcal{G}$ we have*

$$\operatorname{codim}(z, x) = \operatorname{codim}(z, y) + \operatorname{codim}(y, x)$$

When $\mathcal{G}$ is irreducible this is equivalent to

$$\operatorname{codim}(z) = \operatorname{codim}(z, y) + \operatorname{codim}(y)$$

for all $z < y$.

Proof. Suppose $\mathcal{G}$ is catenary. Choose saturated chains C_1 in $[z, y]$ and C_2 in $[y, x]$. One may show that $C_1 \cap C_2 = \{y\}$ and $C_1 \cup C_2$ is a saturated chain in $[x, z]$. The result follows by definition of catenary.

Conversely, consider a saturated chain between x and y of length n

$$x = x_0 \lneq x_1 \lneq ... \lneq x_n = y$$

Then clearly $\operatorname{codim}(x_i, x_{i+1}) = 1$ for all $i = 0...n-1$. Therefore by repeatedly applying the relation we may show that

$$\operatorname{codim}(x, y) = n$$

and therefore every saturated chain between x and y has the same length.

When $\mathcal{G}$ is irreducible we may deduce the second formula by setting $x = \top$. Conversely we may see that

$$\operatorname{codim}(z, x) = \operatorname{codim}(z) - \operatorname{codim}(x) = \operatorname{codim}(z) - \operatorname{codim}(y) + \operatorname{codim}(y) - \operatorname{codim}(x) = \operatorname{codim}(z, y) + \operatorname{codim}(y, x)$$

■

Lemma 2.5.10
Let $\mathcal{G}$ be a biequidimensional finite-dimensional poset. Then for $x \in \mathcal{G}$ we have

a) $\{x\}^{\downarrow}$ and $\{x\}^{\uparrow}$ are biequidimensional
b) $\dim(\mathcal{G}) = \dim(x) + \operatorname{codim}(x)$
c) If x is maximal then $\dim(x) = \dim(\mathcal{G})$ and in particular $\mathcal{G}$ is equidimensional
d) If x is minimal then $\operatorname{codim}(x) = \dim(\mathcal{G})$ and in particular $\mathcal{G}$ is equicodimensional

In particular $\mathcal{G}$ is quasi-biequidimensional.

Proof. Consider a fixed maximal chain C of $\{x\}^{\uparrow}$, necessarily containing x. Any maximal chain C' of $\{x\}^{\downarrow}$ also contains x and combines with C to yield a maximal chain of $\mathcal{F}$. Whence $\ell(C') + \ell(C) = \dim(\mathcal{F})$ and $\{x\}^{\downarrow}$ is biequidimensional. By duality $\{x\}^{\uparrow}$ is biequidimensional and $\ell(C) = \operatorname{codim}(x)$ from which the formula follows.

If x is maximal then clearly $\operatorname{codim}(x) = 0$, and similarly if x is minimal then $\dim(x) = 0$, so the last two statements follow immediately. ■

Proposition 2.5.11 (Equivalent Characterizations of Biequidimensionality)
Let $\mathcal{G}$ be a finite-dimensional poset. Then the following are equivalent

a) $\mathcal{G}$ is quasi-biequidimensional
b) $\mathcal{G}$ is catenary and for every maximal x we have $\{x\}^{\downarrow}$ is equicodimensional
c) $\mathcal{G}$ satisfies $\dim(x) = \dim(y) + \operatorname{codim}(y, x)$ for $y \leq x$
d) $\mathcal{G}$ satisfies c) whenever $\operatorname{codim}(y, x) = 1$

Furthermore the following relationship holds

$$\operatorname{codim}(y) = \operatorname{codim}(y, x) + \operatorname{codim}(x) \quad y \leq x$$

Proof. a) $\implies$ c). By assumption $\{x\}^{\downarrow}$ is biequidimensional. Then

$$\dim(x) = \dim\big(\{x\}^{\downarrow}\big) \overset{(2.5.10)}{=} \dim\big(y; \{x\}^{\downarrow}\big) + \operatorname{codim}\big(y; \{x\}^{\downarrow}\big) = \dim(y) + \operatorname{codim}(y, x)$$

For c) $\implies$ b). Suppose $z < y < x$ in $\mathcal{G}$ then by the codimension formula applied twice

$$\operatorname{codim}(z, x) = \dim(x) - \dim(z) = \dim(x) - \dim(y) + \dim(y) - \dim(z) = \operatorname{codim}(y, x) + \operatorname{codim}(z, y)$$

so by (2.5.9) $\mathcal{G}$ is catenary. Let x be a maximal element and z a minimal element of $\{x\}^{\downarrow}$, then by the codimension formula

$$\operatorname{codim}(z, x) = \dim(x) - \dim(z) = \dim(x)$$

whence $\{x\}^{\downarrow}$ is equicodimensional.

b) $\implies$ a). Let x be a maximal element and C a maximal chain in $\{x\}^{\downarrow}$ with minimum element y then, as $\mathcal{G}$ is catenary, we have $\ell(C) = \operatorname{codim}(y, x)$. As $\{x\}^{\downarrow}$ is equicodimensional we have $\operatorname{codim}(y, x) = \dim(x)$ which shows that $\{x\}^{\downarrow}$ is biequidimensional.

Clearly c) $\implies$ d). Conversely for d) $\implies$ a) consider a maximal chain in $\{x\}^{\downarrow}$

$$x_0 \lneq \ldots \lneq x_n = x$$

Then clearly $\operatorname{codim}(x_i, x_{i+1}) = 1$ whence by induction $\ell(C) = \dim(x)$.

The final statement follows from (2.5.9) because $\{x\}^{\downarrow}$ is irreducible and catenary. ■

Remark 2.5.12
This is a corrected version of EGA IV 14.3.3, as noted by Heinrich.

Corollary 2.5.13
When $\mathcal{F}$ is a quasi-biequidimensional Krull Lattice then we have the following codimension formulas for all $x \in \mathcal{J}(\mathcal{F})$ and $y \in \mathcal{F}$ such that $y \leq x$

$$\dim(x) = \dim(y) + \operatorname{codim}(y, x)$$
$$\operatorname{codim}(y) = \operatorname{codim}(y, x) + \operatorname{codim}(x)$$

Furthermore if $\mathcal{F}$ is biequidimensional then for all $y \in \mathcal{F}$

$$\dim(\mathcal{F}) = \dim(y) + \operatorname{codim}(y)$$

Proof. The first two relations hold when $x, y \in \mathcal{J}(\mathcal{F})$ by (2.5.11). We may generalise this as follows.

For the first relation and fixed $x \in \mathcal{J}(\mathcal{F})$ we see $\dim(y_j)$ is maximal precisely when $\operatorname{codim}(y_j, x)$ is minimal. Therefore it holds for all $y \in \mathcal{F}$.

Similarly for the second relation we see that $\operatorname{codim}(y_j, x)$ is minimal precisely when $\operatorname{codim}(y_j)$ is minimal.

The final statement follows from the first relation by taking the supremum over all maximal elements x. ■

Corollary 2.5.14
Let $\mathcal{G}$ be an irreducible finite-dimensional poset. Then the following are equivalent

a) $\mathcal{G}$ is biequidimensional
b) $\mathcal{G}$ is quasi-biequidimensional
c) $\mathcal{G}$ is catenary and equicodimensional
d) $\dim(x) = \dim(y) + \operatorname{codim}(y, x) \quad \forall y \leq x$
e) $\mathcal{G}$ satisfies d) when $\operatorname{codim}(y, x) = 1$

Proof. The equivalence follows from (2.5.11) by noting that an irreducible poset has only one maximal element and is in particular equidimensional. ■

Proposition 2.5.15
Let $\mathcal{F}$ be a quasi-biequidimensional Krull Lattice. Then for all $x \in \mathcal{F}$ we have $\{x\}^{\downarrow}$ is a quasi-biequidimensional Krull Lattice.

Proof. Consider $y \in \mathcal{J}(\{x\}^{\downarrow}) \overset{(2.4.3)}{=} \mathcal{J}(\mathcal{F}) \cap \{x\}^{\downarrow}$. Then by definition $\{y\}^{\downarrow}$ is biequidimensional, and so $\{x\}^{\downarrow}$ is quasi-biequidimensional. ■